Core: Required for any one of the four levels of certification.

Stratospheric Ozone Depletion

Stratospheric ozone helps form the earth's protective shield. The ozone layer protects the earth from ultraviolet radiation from the sun. Ozone depletion potential (or ODP) measures the ability of a substance to destroy ozone in the stratosphere. Ozone depletion in the stratosphere is a global problem which affects everyone on the planet. One of the most serious results from the thinning and damaging of the stratospheric ozone layer is an increase in the rates of skin cancer in humans. Another human health effect that has increased from damage to the stratospheric ozone layer is the increase in cases of cataracts. Stratospheric ozone depletion can also have an effect on the environment, causing decreased crop yields and damage to marine organisms.

There has been some controversy over the subject of ozone depletion. Some believed that the chlorine found in the stratosphere comes from natural sources such as volcanic eruptions. However, air samples taken over erupting volcanoes show that volcanoes add only small quantities of chlorine to the atmosphere compared to the amount of chlorine added to the atmosphere from chlorine-containing refrigerants. In addition, the rise in the amount of chlorine measured in the stratosphere over the past four decades matches the rise in the amount of Fluorine, which has different natural sources than chlorine over the same period. Also, the rise in the amount of chlorine measured by NASA and other agencies in the stratosphere over the past twenty years matches the rise in CFC and HCFC emissions over the same period. The evidence is clear, chlorine containing refrigerants have changed the natural balance, thus depleting the ozone layer.

Ozone in the stratosphere consists of molecules containing 3 oxygen atoms, O_3. Chlorine is the element of a CFC or HCFC molecule that causes ozone depletion. When a chlorine atom encounters an ozone molecule, it takes one of the ozone's 3 oxygen atoms, forming a compound called chlorine monoxide (ClO), and leaves an oxygen O_2 molecule behind, Figure 5. The chlorine monoxide will collide with another ozone molecule, releasing its Oxygen atom, forming two O_2 molecules, leaving the chlorine free to attack another ozone molecule. A single chlorine atom can last in the Stratosphere for 120 years destroying up to 100,000 ozone molecules.

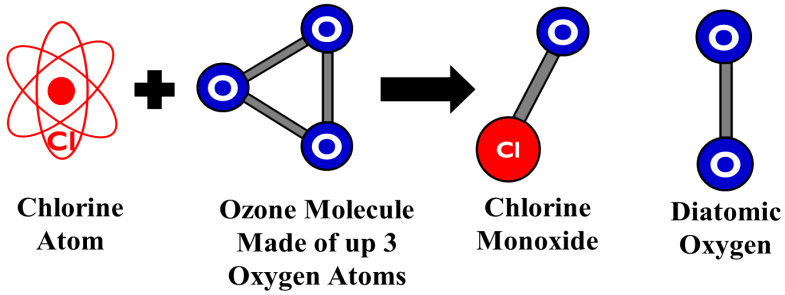

Figure 5. Chlorine / Ozone.

Unlike other chlorine compounds and naturally-occurring chlorine, the chlorine in CFCs and HCFCs will neither dissolve in water nor break down into compounds that dissolve in water so they do not rain out of the atmosphere.

HFCs are made up of Hydrogen, Fluorine, and Carbon. They do not contain chlorine that effects the ozone layer, but most HFCs have a high Global Warming Potential (GWP).

Global Warming Potential

Global Warming Potential (GWP) was established to provide comparisons of the global warming impacts of different gases over a span of time. Carbon Dioxide (CO_2) is used as the baseline measurement for global warming potential and has a value of 1. Most HFC refrigerants, such as R-410A, have global warming potentials thousands of times greater compared to carbon dioxide. Hydrocarbon (HC) and Hydrofluoroolefin (HFO) refrigerants have the lowest global warming potential compared to CFCs, HCFCs, and HFCs.

The global warming potential of isobutane (R-600a), propane (R-290), and R-441A are significantly lower when compared to the GWPs of HFCs such as R-134a, R-407C, R-404A, and R-410A.

With non-ozone depleting and very low global warming potential characteristics, hydrofluoroolefin (HFO) refrigerants are a good fit for HVACR equipment. Most HFCs have very high global warming potentials; a characteristic that makes them damaging to the environment.

Refrigerant Characteristics & Identification

Refrigerants may be identified by type (CFC, HCFC, HFC, HFO, HC) and American Society of Heating Refrigerating and Air-Conditioning Engineers (ASHRAE) number or with "R" plus the ASHRAE number such as HFC-134a or R-134a.

ASHRAE has grouped refrigerants by Class A (safest) or B, depending on their toxicity level to humans. Flammability is indicated by a 1 (no flammability), 2 (low flammability), or 3 (high flammability), Figure 6.

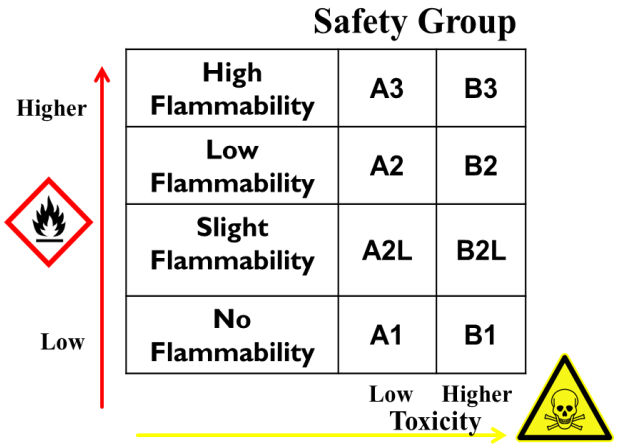

Figure 6. ASHRAE Safety Group.

Chlorofluorocarbon (CFC) refrigerants have the highest ozone depletion potential (ODP) and are the most harmful to stratospheric ozone. CFC refrigerants contain chlorine, fluorine, and carbon. Chlorine is the component that poses a threat to the stratospheric ozone. Examples of CFC refrigerants include R-11 and R-12.

Hydrochlorofluorocarbon (HCFC) refrigerants contain hydrogen, chlorine, fluorine, and carbon. The hydrogen makes them a little less harmful to the stratospheric ozone when compared to CFCs. R-22 and R-123 are examples of HCFC refrigerants that have an ozone depletion potential of more than zero, ranging between zero and one.

Hydrofluorocarbon (HFC) refrigerants contain hydrogen, fluorine, and carbon atoms connected by single bonds between the atoms. HFCs have the lowest/no ozone depletion potential but do have an effect on global warming. HFC refrigerants do not contain harmful chlorine. Examples of HFCs are R-134a, R-410A, R-404A, R-407C, and R-422B.

Hydrofluoroolefin (HFO) refrigerants contain hydrogen, fluorine, and carbon atoms connected with at least one double bond between the carbon atoms. HFOs have no ozone depletion potential but do have a very small effect on global warming. HFOs are less flammable than hydrocarbon refrigerants but most are still mildly flammable so are classified as A2L. HFO refrigerants contain fluorine making them less flammable than hydrocarbon refrigerants. HFO refrigerants are miscible in POE lubricants. R-1234yf is an HFO refrigerant used in chillers.

Hydrocarbon (HC) refrigerants are an elementary compound of hydrogen and carbon. Although flammable, these refrigerants pose the least amount of danger to the environment. HCs have no ozone depletion potential and a very small effect on global warming, with GWP values less than 10. Isobutane (R-600a) and Propane (R-290) are examples of HCs. Some HC refrigerants are considered natural refrigerants.

Note: Propane cylinders for grilling should never be used as a refrigerant, as they contain impurities that can damage refrigeration equipment.

Azeotropic refrigerant mixtures contain two or more refrigerants. At a certain pressure, an azeotropic mixture evaporates and condenses at a constant temperature. It acts like a single component refrigerant over its entire range. Thus, azeotropic mixtures act as a pure compound.

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Zeotropic or **non-azeotropic** refrigerant mixtures are mixtures of components that have different boiling points. This mixture is a blend of refrigerant. The blend may be binary (two-part) or ternary (three-part). Blended refrigerants are normally associated with having what is called "temperature glide". This is due to the blended parts having different pressures for the same saturation temperature. Temperature glide can range a few tenths of a degree to 12 degrees or more.

Bubble and Dew Points

Zeotropic and some near-azeotropic blends use bubble and dew points to indicate condensing and evaporation temperatures on a pressure-temperature chart. Bubble point (liquid) values are used when charging by condenser subcooling. Dew point (vapor) values are used for charging by suction or evaporator superheat. The relationship of pressures and temperatures are based on the refrigerant's temperature glide.

The R-400 series of refrigerants are near azeotropic blends that can leak from a system at uneven rates due to different vapor pressures which can affect the percentage of each refrigerant remaining in the system. The proper charging method for R-400 series blended refrigerants is to weigh the refrigerant into the high side of the system as a liquid. When adding refrigerant to an undercharged system, liquid refrigerant is throttled into the low side with the system operating.

There is concern for using blended refrigerants in systems that develop a leak. There may be fractionization of the refrigerant (refrigerant leaking at uneven rates due to different vapor pressures). R-407C is one of the refrigerants that has a high-temperature glide and may easily experience fractionation. Even though the pressure-temperature relationship and operating characteristics may be almost the same, the refrigerants cannot be interchanged. Although the saturation pressure-temperature behavior of two refrigerants might be similar, potential of the refrigerants to oxidize copper tubing must also be considered. Due to flammability of some HCs and HFOs, equipment must be designed to handle the refrigerant. Therefore, EPA regulates the maximum amount of refrigerant that can be used in the new systems by their type and use.

When retrofitting or converting a system to use a different refrigerant, only EPA-approved substitutes can be used. Remember, EPA does not approve any substitute refrigerants as direct drop-in replacements to service an R-22 or any other system.

Low, Medium, High, and Very High Pressure

To better define leak rates and service recovery levels the EPA lists refrigerants in four liquid-phase pressure ranges at 104°F condensing temperature.

Low-pressure refrigerants have a pressure of 30 psig or lower at a liquid-phase temperature of 104°F.
Examples: CFC-11, HCFC-123, HFC-245fa, HFO-1233zd

Medium-pressure refrigerants have a pressure between 30 psig & 155 psig at a liquid-phase temperature of 104°F.
Examples: CFC-12, HCFC-124, HCFC-409A, HFC-134a, HC-600a, HFO-1234yf, HFO-1234ze

High-pressure refrigerants have a pressure between 155 psig & 340 psig at a liquid-phase temperature of 104°F.
Examples: HCFC-22, HFC-404A, HFC-407C, HFC-422B, HFC-422D, HFC-410A, HC-441A, R-717

Very high-pressure refrigerants have a pressure over 340 psig at a liquid-phase temperature of 104°F.
Examples: CO_2 (R-744)

Refrigerant Oils

Most HCFC binary (two-part) and ternary (three-part) blends use a synthetic alkylbenzene lubricant. When retrofitting HCFC equipment to HFC refrigerants or when working with newly manufactured systems, synthetic polyolester oil is most commonly used. Ester-based oils cannot be mixed with any other type of oil. Before retrofitting an HCFC system to HFC refrigerant, the other types of oils must be removed from the system before adding the ester-based oil. Care must be taken when using ester-based oils; they are very hygroscopic and will absorb moisture out of the air and through some containers.

Montreal Protocol and Clean Air Act

The United States is part of the Montreal Protocol which is an international treaty that addresses ozone-depleting substances and their alternatives. To stop damage to the stratospheric ozone layer, the U.S. is phasing out, or has already phased out, the use of chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), and other refrigerants that are harmful to the environment.

Section 608 of the Clean Air Act sets forth the rules and regulations for refrigerant handling in the HVACR industry. Section 608 certification allows a technician to purchase or handle regulated refrigerants. Distributors that sell HCFC, HFC, or another regulated refrigerant must verify the purchaser is, or employs, a Section 608 certified technician.

Anyone performing any activity that violates the refrigerant circuit, including removal of refrigerant or replacing a component containing a regulated refrigerant must be certified at the proper level and have the type of recovery equipment required. It is illegal to knowingly release CFC, HCFC, or HFC refrigerants during the service, maintenance, repair, or disposal of appliances. A technician must have their Section 608 certification card in their possession when preforming an activity regulated by Section 608. If a technician's Section 608 certification card is lost, a replacement can be requested from their certifying organization, not the EPA.

A person is not required to be a Section 608 certified technician when servicing an external electric circuit or sections of the system that do not involve the refrigerant components of an HVACR appliance.

The Clean Air Act (EPA's regulations) allows for releasing an exempt refrigerant (e.g., ammonia and CO_2) or a small amount of refrigerant, mixed with nitrogen (a leak trace gas) for leak detection. As of 2017, a technician can be fined $44,539 per day, per violation, for violating the Clean Air Act, including knowingly releasing non-exempt refrigerant from appliances. Service technicians who violate Clean Air Act provisions may lose their certification in addition to paying fines and being required to appear in court. It is the technician's responsibility to comply with any future changes in the law or EPA regulations after a technician becomes certified. State and local governments may establish laws that contain stricter requirements than the Clean Air Act/EPA regulations.

The EPA's Significant New Alternatives Policy (SNAP) Program identifies refrigerants with lower overall risks to human health and the environment.

The phase-out for the production and imports of newly manufactured R-22 and HCFC-142b in the U.S. will be in the year 2020. After the phase-out of all HCFC refrigerants in 2030, supplies of those refrigerants for equipment servicing will come from recovered and reclaimed gas. It is not a violation of the refrigerant management regulations under the Clean Air Act to service a system that uses CFCs or HCFCs after the phase-out of those refrigerants.

Reclaimed refrigerant must meet AHRI Standard 700 before it can be resold under EPA's regulations. Technicians can only charge used CFC, HCFC, or HFC refrigerants back into the same appliance or into another appliance owned by the same owner. Used refrigerant may no longer meet the AHRI Standard for virgin refrigerant so it cannot change ownership without being reclaimed.

All CFC, HCFC, HFC, and HFO refrigerant recovery equipment must meet EPA standards. Recovery equipment must be tested according to AHRI Standard 740. Recovery machines may be able to recover one or multiple types of refrigerants.

Desiccant dehumidifiers are NOT covered by EPA Section 608 regulations as they do not contain a refrigerant.

Self-contained recovery devices remove refrigerant from an appliance without the assistance of components in the appliance. Recovery equipment that relies on the compressor in the appliance and/or the pressure of the refrigerant in the appliance is considered to be system-dependent.

To make it easier to recover refrigerant, EPA regulations require appliances that do not have service valves to have a service aperture or process stub.

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When disposing of appliances with 5-50 pounds of refrigerant, EPA does not require technicians (or the company employing the technician) to record the model and serial number of the appliance. However, records that must be kept include;

- type of refrigerant recovered
- quantity of refrigerant, by type
- quantity recovered from disposed appliances in each calendar month
- quantity of refrigerant, by type, sent for reclamation or destruction.

EPA's refrigerant management regulations exempt certain refrigerants from the venting prohibition. This can occur when and if the EPA determines that a certain refrigerant does not pose a threat to the environment if released. Some natural refrigerants, such as carbon dioxide, can be purchased by someone who is not Section 608 certified.

It is lawful to service existing R-22 systems and use existing supplies of R-22. It is not legal to top off an R-22 charge with another refrigerant such as R-410A. Topping off one type of refrigerant with another can void equipment warranty, ruin the refrigerant, damage the equipment, and cause severe personal injury, or death.

When discarding any disposable cylinders of CFCs, HCFCs, HFCs, or HFOs, recover all remaining refrigerant, render the cylinder useless, and recycle the metal.

Before opening or disposing of any appliance containing a chlorofluorocarbon (CFC), hydrochlorofluorocarbon (HCFC), or hydrofluorocarbon (HFC) refrigerant, recover the refrigerant to the approved EPA recovery or evacuation levels. Under the Section 608 "Safe Disposal Requirements", it is the final person in the disposal chain that is responsible for ensuring that any CFC, HCFC, or HFC refrigerant has been removed from household refrigerators or other appliances before they are disposed of.

The knowingly release of a CFC, HCFC, HFC or HFO refrigerant during the service, maintenance, repair, vandalism, theft, or disposal of appliances is illegal. Refrigerants emitted when cutting a line without properly evacuating the appliance violates the venting prohibition. Venting refrigerant from a recovery machine or recovery cylinder after use is also illegal. Records of recovered refrigerant must be kept to ensure venting does not take place after recovery from an appliance.

Small quantities of isobutane, used in household freezers, can be vented. Releasing mixtures of nitrogen and refrigerant that result from adding nitrogen to a fully charged appliance to leak check the appliance is considered a violation of the prohibition on venting. Only a few ounces of refrigerant mixed with nitrogen is considered a leak trace gas and may be released without recovery.

Recovery

The processes of recovery, recycling, and reclaiming sound similar, but they are quite different.

Recover means to remove refrigerant, in any condition from a system and store it in an approved recovery cylinder or container. Recovered refrigerant may be charged into the same system, another system with the same ownership, or shipped to an approved reclaimer.

Recycle means to extract refrigerant from an appliance (except motor vehicle air conditioners or MVACs) and clean the refrigerant for reuse in equipment of the same owner without meeting all of the requirements for reclamation. In general, recycled refrigerant is refrigerant that is cleaned using oil separation and single or multiple passes through devices, such as replaceable core filter-driers, which reduce moisture, acidity, and particulate matter. Recycled refrigerant may be charged into the same system or another system with the same ownership. Recycling equipment must meet AHRI 740 Performance Rating Standard.

Reclaim means to process refrigerant to a level equal to new (virgin) product specifications as determined by chemical analysis. Reclaimed refrigerants must meet AHRI 700 Standards, before it can be sold. This is normally preformed at a chemical processing company.

Recovery Devices

Refrigerant recovery and/or recycling equipment manufactured after November 15, 1993 must be certified and labeled by an EPA-approved equipment testing organization to meet EPA standards. An EPA-approved certification label is required on all new recovery equipment.

There are two basic types of recovery devices: “System-dependent” (passive) which captures refrigerant with the assistance of components in the appliance from which refrigerant is being recovered, Figure 7. "Self-contained” (active) which has its own means to draw the refrigerant out of the appliance, Figure 8.

Figure 7. Passive Recovery Bag 12 oz or less.

Figure 8. Active Recovery Unit. Courtesy of Appion Inc.

The recovery time will be increased if the appliance is located in low ambient temperatures or long hoses are used between the unit and the recovery machine. Long and small diameter hoses between the unit and recovery machine should be avoided as they will cause excessive pressure drop and increase recovery time. Upon completion of refrigerant liquid transfer between recovery unit and the refrigeration system, guard against trapping liquid refrigerant in the service hose between closed service valves.

The technician must have a separate refrigerant recovery cylinder for each type of refrigerant recovered and should have a separate cylinder for refrigerants known to be mixed. If servicing systems that use HFC-134a, HFC-410A and HCFC-22, then three cylinders are required even though two are HFCs. It is important not to mix refrigerants in the same container when recovering as refrigerants that are mixed be may be impossible to reclaim and there may be an added cost to have them destroyed.

Technicians can only charge used CFC, HCFC, or HFC refrigerants back into the same appliance or into another appliance with the same owner. Used refrigerant may no longer meet the AHRI standard for virgin refrigerant so it cannot change ownership without being reclaimed.

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Leak Detection

Leak detection methods and devices used must be approved by the manufacturer. The wrong type of electronic leak detector used with a flammable refrigerant can cause an explosion and bodily injuries.

To determine the general area of a small leak, the use of an electronic or ultrasonic leak detector is considered to be the most effective. To cause the least amount of damage to the environment, newly installed systems should be pressurized with dry-nitrogen for leak detection purposes. If an electronic leak detector is to be used, a trace of the system refrigerant can be added to the system along with the dry-nitrogen. The trace gas should be the same as the refrigerant the appliance is going to be charged with. A leak trace gas is not considered a refrigerant under the EPA's refrigerant management regulations. Once the general area of the leak is located, the use of soap bubbles will aid in pinpointing the leak. Finding and repairing leaks in a system will conserve refrigerant for future use.

Upon completion of a leak repair, and before recharging the system, install a new filter drier, and complete a standing-pressure leak check at the maximum system pressure. For safety, any system designed to contain a flammable hydrocarbon (HC) or HFO refrigerant has to have a leak test before evacuating to 500 microns or lower.

Unlike natural gas, flammable hydrocarbon or HFO refrigerants do not contain odorants to indicate their presence. Odorants do not need to be added to systems containing flammable refrigerants.

Dehydration

The reason for dehydrating a refrigeration system is to remove water and water vapor, and it is important to follow proper dehydration procedures. If moisture is allowed to remain in an operating refrigeration system, hydrochloric and hydrofluoric acids may form. To dehydrate a system, it should be evacuated. It is not possible to over-evacuate a system. Most manufacturers require system evacuation to 500 microns, or lower, which can be measured with a micron gauge, Figure 9.

Figure 9. Micron Gauge. Courtesy of Appion Inc.

Never evacuate a system to the ambient air without first following proper recovery procedures and attaining the mandated recovery level required by EPA regulations.

The factors that affect the speed and efficiency of evacuation are; size of equipment being evacuated, ambient temperature, amount of moisture in the system, the size of the vacuum pump and vacuum lines, Figures 10 & 11. In addition, vacuum lines (hoses) should be equal to or larger than the pump intake connection. The brand of a micron gauge does not usually affect the speed of evacuation, but the method used to connect it to the system may. The piping connection to the vacuum pump should be as short in length as possible and as large in diameter as possible. For accurate readings during evacuation, the system vacuum gauge should be connected as far away from the vacuum pump as possible. Measuring a final system vacuum should be done with the system isolated and the vacuum pump turned off. A system that will not hold a vacuum after it has been evacuated probably has a leak. During evacuation, you may wish to heat the refrigeration system to decrease dehydration time. Tapping a compressor with a rubber mallet will aid in releasing trapped refrigerant from the oil.

Figure 10. Vacuum Pump. Courtesy of Appion Inc.

Figure 11. Vacuum Manifold. Courtesy of Appion Inc.

Dehydration is complete when the vacuum gauge/indicator shows that you have reached and held the required vacuum of 500 microns or less. If the pressure increases then stops for a few minutes then rises more, there is probably moisture left in the system. If the pressure steadily rises, there is probably a leak.

Safety/General

The EPA is concerned not only with the prevention of refrigerant venting, but with the technician's overall safety. When handling and filling refrigerant cylinders or operating recovery or recycling equipment, you should wear safety glasses, protective gloves, and follow all equipment manufacturer's safety precautions. Make sure your recovery machine is grounded when in use, especially when recovering a flammable refrigerant.

When pressurizing a system with nitrogen, you should always charge through a pressure regulator and a relief valve in the downstream line from the pressure regulator. When using more than one pressure relief valve, they MUST NOT be installed in series TO EACH OTHER, only parallel to each other. If corrosion build-up is found within the body of a relief valve, the valve MUST be replaced.

When leak checking a system, NEVER pressurize the system with oxygen or compressed air. When mixed with some refrigerants or compressor oil, oxygen or compressed air can cause an explosion. To determine the safe pressure for leak testing, check the equipment data plate for the maximum low-side test pressure value.

NEVER expose refrigerants to open flames or glowing hot metal surfaces. Although reclaimers may accept visibly burned recovery tanks, at high temperatures, refrigerants decompose and form acids. Hydrochloric acid is formed if the refrigerant contains chlorine. Hydrofluoric acid is formed if the refrigerant contains fluorine. Heating a refrigerant cylinder can result in an explosion and cause serious injuries. If oxygen is also present, it is possible to form carbon monoxide, carbon dioxide, and phosgene gas. If a hydrocarbon (HC) refrigerant is released into a space, and an ignition source is provided an explosion can occur if the refrigerant concentration is above the Lower Flammability Limit (LFL) and below the Upper Flammability Limit (UFL) with an ignition source.

When working with any solvents, chemicals, or refrigerants, Safety Data Sheets (SDS) should be reviewed by the technician to reference compatibility with other materials. Silicone elastomers, for example, which are used in seals and gaskets, are not compatible with HFO refrigerants.

Refrigerant vapors, or mist, in high concentrations, should not be inhaled; they may cause heart irregularities or unconsciousness. In most refrigerant accidents where death occurs, oxygen deprivation is the major cause. According to the ASHRAE refrigerant safety classification standard, A1 designations would be the safest. R-410A is considered to be non-toxic but it, as any other refrigerant in high concentrations, can cause asphyxia.

Prior to recovery of any flammable refrigerant ensure that the refrigeration system and recovery unit is grounded. Never apply an open flame or live steam to a flammable refrigerant cylinder, do not cut or weld any refrigerant line while refrigerant is in the unit, and do not use oxygen to purge lines or to pressurize the machine. A sufficient concentration of the refrigerant in the space and an ignition source are most likely to lead to an explosion with refrigerants. HFO-1234yf is classified A2L not as an A3 (higher flammability) refrigerant like Isobutane (R-600a), Propane (R-290), and R-441A (Hydrocarbon blend).

A red color marking is required on all process tubes and other pipes through which a flammable refrigerant flows/passes and where a service connection is probable. The red marking must extend a minimum of one inch in both directions from such locations.

In the event of a large release of CFC, HCFC, HFC, HFO, HC, or any refrigerant in a contained area, a self-contained breathing apparatus (SCBA) or leaving the area is required.

Note: An alcohol spray should be used to remove ice from sight glasses or viewing glasses.

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Safety/Cylinder

A refrigerant recovery cylinder should be free of rust, damage, be labeled, secure, and filled to no more than 80% of its capacity by weight. Approved refrigerant recovery cylinders can be identified by having yellow colored tops and gray bodies, Figure 12. One should not heat a refrigerant storage or recovery tank with an open flame because refrigerant in the tank may decompose forming a toxic material, it can result in venting refrigerant to the atmosphere from the pressure safety valve or the tank may explode, causing serious injury.

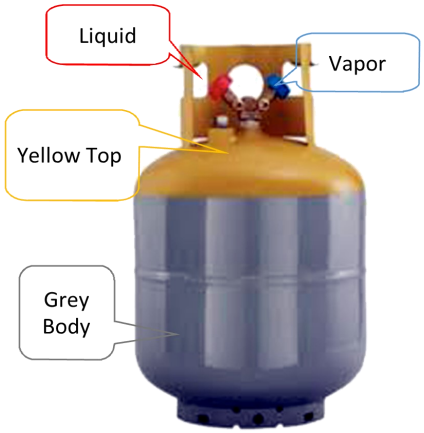

Figure 12. Recovery Tank.

When using vapor or liquid recovery, the fill level of the recovery cylinder can be controlled by a mechanical float device, electronic shut-off device, or measuring the gross cylinder weight. The pitch of the recovery device does not control the fill level of the recovery cylinder. Care must be used to prevent overfilling refrigerant storage cylinders. The internal pressure of the cylinders may rise in heated areas and cause an explosion. A reusable container for refrigerant that is under high pressure (above 15 psig), at normal ambient temperature, must be hydrostatically tested every 5 years.

A disposable refrigerant cylinder may **never** be used to recover refrigerant. When empty, the vapor should be recovered, reducing the pressure in a disposable cylinder to 0 psig before scrapping.

Shipping

When transporting cylinders containing used refrigerant, a Department of Transportation (DOT) classification tag and label must be attached to the cylinders. Portable, refillable tanks or containers used to ship CFC, HCFC, or other refrigerants obtained with recovery equipment must meet DOT standards. DOT regulations require that the number of cylinders of each gas be recorded on the shipping paper for hazard class 2.2 Nonflammable Compressed Gases. Refrigerant cylinders must be positioned upright when they are shipped.

Before shipping any used refrigerant in a cylinder, a refrigerant label to identify the type of refrigerant recovered is placed on a recovery cylinder to avoid accidental mixing of recovered refrigerants, allow the recycler to identify the contents, and allow the technician's company to determine the amount of refrigerant recovered for recordkeeping purposes.